

The quadratic trace is deterministic: a disproof of Conjecture 00000007744

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Abstract

In the uniform simple d -regular graph model, the squared adjacency matrix has trace nd for every graph on n vertices. Thus its centered quadratic trace, with any deterministic matrix and trace normalization and with the stated $\sqrt{n}$ factor, is identically zero. Its law is exactly δ_0 at every admissible size, and every weak limit has variance zero. For $d = 3$, the conjecture instead prescribes variance $8/9$. We formalize the actual uniform model, polynomial traces, finite-dimensional laws, and the weak-limit contradiction in Lean 4.19.0 with Mathlib v4.19.0.

1 The original claim and the clause refuted

The supplied English statement reads:

Definition: The spectral empirical measure of random d -regular graphs (uniform model) converges to the Kesten-McKay law KM_d ; the second-order limit is the joint limit process of normalized trace polynomials times $\sqrt{n}$. Conjecture: The limit is Gaussian with covariance $\Sigma(p, q)$ given by $(d + 1)$ -regular constrained counts of noncrossing return path pairings of p and q ; for $p = q = x^2$ one has $\text{Var} = 4 - 12/d + 8/d^2$, degenerating to the Wigner value 4 of GUE fluctuations as d grows. (Kesten-McKay second-order fluctuations)

The Chinese version makes the centering explicit:

定义: 随机 d -正则图 (一致模型) 的谱经验测度收敛到 Kesten-McKay 律 KM_d ; 其二阶极限分布指规范化迹多项式 $(\text{tr } p(A) - \mathbb{E}) \cdot \sqrt{n}$ 的联合极限过程。猜想: 该极限为高斯族, 协方差函数 $\Sigma(p, q)$ 由 p, q 的非交叉返回路径配对的 $(d + 1)$ -正则约束计数给出; 特别对 $p = q = x^2$, $\text{Var} = 4 - 12/d + 8/d^2$, 且当 $d \rightarrow \infty$ 时退化为 GUE 波动的 Wigner 值 4。

Both versions require the displayed quadratic-coordinate variance. We disprove that explicit requirement at fixed degree 3. We do not dispute the first-order Kesten-McKay convergence or claim that every other coordinate vanishes. No interpretation of the unspecified general pairing rule is needed. The exact original Markdown file is included as `original_conjecture.md`.

2 The whole uniform model and its quadratic trace

We use the conventional uniform model of finite labelled *simple*, undirected d -regular graphs. Thus loops and parallel edges are absent; we do not assert a result about an unconditioned multigraph configuration model. The source states no connectedness requirement. On any finite label set V of size n , let

$$\mathcal{G}_{V,d} = \{G : G \text{ is a simple graph on } V, \deg_G(v) = d \text{ for every } v \in V\}.$$

When this set is nonempty, give *every* graph mass $1/|\mathcal{G}_{V,d}|$. The Lean definitions use the subtype of `SimpleGraph V` satisfying `IsRegularOfDegree d`, and `PMF.uniformOfFintype` on that entire

subtype. An explicit theorem verifies the equal masses. The measurable space consists of all subsets.

Write A_G for the 0–1 adjacency matrix. Let $a, b \in \mathbb{R}$ be deterministic scalars and put

$$T_{a,b}(G) = b \operatorname{Tr}((aA_G)^2), \quad Y_{a,b}(G) = (T_{a,b}(G) - \mathbb{E}T_{a,b})\sqrt{n}.$$

Ordinary adjacency and empirical normalized trace correspond to $a = 1$ and $b = 1/n$. Allowing arbitrary a, b also covers ordinary unnormalized trace and deterministic rescalings of the adjacency matrix.

Proposition 1. *For every graph in $\mathcal{G}_{V,d}$,*

$$T_{a,b}(G) = ba^2nd, \quad \mathbb{E}T_{a,b} = ba^2nd, \quad Y_{a,b}(G) = 0.$$

Consequently $\operatorname{Law}(Y_{a,b}) = \delta_0$.

Proof. For each $v \in V$, symmetry and 0–1 adjacency entries give

$$(A_G^2)_{vv} = \sum_{w \in V} (A_G)_{vw}(A_G)_{wv} = \sum_{w \in V} (A_G)_{vw} = \deg_G(v) = d.$$

Summing the diagonal yields $\operatorname{Tr}(A_G^2) = nd$, and scalar multiplication gives the asserted formula for $T_{a,b}$. It is a constant on the *whole* sample space, so its integral against the uniform probability measure is the same constant. Subtracting that expectation gives zero pointwise. Pushing the probability measure forward by this constant map gives δ_0 . $\square$

3 Nonempty large models and the actual weak limit

For each $k \geq 0$ take

$$V_{d,k} = \{0, \dots, k\} \times \{0, \dots, d\}, \quad n_{d,k} = (k+1)(d+1).$$

The graph with (i, u) adjacent to (j, v) exactly when $i = j$ and $u \neq v$ is a disjoint union of $k+1$ copies of K_{d+1} , hence is d -regular. This proves nonemptiness of the sample space at every such size. The random graph is still drawn uniformly from *all* regular graphs on $V_{d,k}$; the block graph is only an existence witness. Lean proves the stated cardinality, its positivity, and $n_{d,k} \rightarrow \infty$.

Let a_k, b_k be any deterministic real sequences and let μ_k be the actual law of Y_{a_k, b_k} on this model. Proposition 1 gives $\mu_k = \delta_0$ for every k , so $\mu_k \Rightarrow \delta_0$. If also $\mu_k \Rightarrow \nu$, uniqueness of weak limits gives $\nu = \delta_0$, and therefore

$$\operatorname{Var}_\nu(\operatorname{id}) = 0.$$

This argument identifies the entire limiting law. It does not interchange variance and weak limits. Lean uses the usual weak topology on $\operatorname{ProbabilityMeasure} \mathbb{R}$, its Hausdorff property, and the ordinary measure-theoretic variance.

For $d = 3$ this is an unbounded sequence of admissible sizes $4(k+1)$. A limit asserted over all admissible sizes must also hold along this sequence. The finite trace identity holds on every finite label set; the product labels impose no additional restriction on the sampled graphs.

4 From the joint polynomial process to the contradiction

For an arbitrary real polynomial p , the formal project defines

$$T_p(G) = b \operatorname{Tr}(p(aA_G)), \quad Z_p(G) = (T_p(G) - \mathbb{E}T_p)\sqrt{n}$$

using `Polynomial.aeval`, the actual matrix trace and the actual uniform integral. These observables are integrable on the finite sample space. It proves that $p = x^2$ specializes to the preceding quadratic observable. For any finite list $p_1, \dots, p_m$, it constructs the genuine joint law of $(Z_{p_1}, \dots, Z_{p_m})$ and identifies each coordinate marginal. Continuous coordinate projection sends a weakly convergent sequence of these joint laws to a weakly convergent sequence of their marginals.

The predicate `RequiredSecondOrderLimit` states a *necessary part* of the original claim: these finite-dimensional laws have weak limits and the one-coordinate list $p = x^2$ has the prescribed limiting variance. Gaussianity and the other covariance prescriptions are further requirements of the original claim. They are not silently redefined in this predicate. The formal disproof of this necessary part is sufficient to refute the original conjunction.

At $d = 3$ the required variance is

$$4 - \frac{12}{3} + \frac{8}{3^2} = \frac{8}{9} \neq 0.$$

It is incompatible with the uniquely forced law δ_0 . Thus no proposed Gaussian family with the source's covariance can be the joint limit. A degenerate Gaussian coordinate would be consistent with zero fluctuations; the contradiction is the explicitly positive variance.

The general formal conclusion is

$$\forall a, b : \mathbb{N} \rightarrow \mathbb{R}, \quad \neg \text{RequiredSecondOrderLimit}(3, a, b),$$

proved by `no_required_second_order_limit`. The final theorem `conjecture7744_false` specializes to $a_k = 1$ and $b_k = 1/n_{3,k}$. No unproved first-order spectral limit or central limit theorem is used.

5 Formal and computational verification

The project pins Lean 4.19.0 and Mathlib v4.19.0 (Mathlib commit `c44e0c8ee63ca166450922a373c7409c5d26b00b`), with all nine dependency revisions recorded in `lake-manifest.json`. The project build treats warnings as errors. The author, coordinator and independent reviewer performed fresh builds. The coordinator and reviewer also replayed every final module with `-DwarningAsError=true`.

There are 49 explicitly authored declarations: 19 definitions, 26 theorems, and four named instances. The full compiled inventory contains 83 declarations: 75 logical declarations and eight compiler-generated runtime wrappers. Every logical declaration has only the standard Lean axioms `propext`, `Classical.choice`, and `Quot.sound`, or none. There are no axiom declarations, admissions, `native_decide`, partial logical declarations, or unsafe/missing logical dependencies. The runtime wrappers are retained in the inventory and are absent from logical dependency closures. The author source has no unsafe proof mechanism.

The supplied exact-arithmetic Python check enumerates every labelled simple cubic graph on four and six vertices, computes the full matrix square, and then computes uniform expectations and centered variances:

n	Graph count	$\text{Tr}(A^2)/n$	Centered variance	Claimed limit variance
4	1	3	0	8/9
6	70	3	0	8/9

The reviewer independently replayed that script and an alternative complete edge-mask enumeration, with matching results. These finite calculations are supporting checks only; the asymptotic disproof is the general Lean argument.

The accompanying README gives reproduction commands and the evidence map. `Audit.lean` prints axiom dependencies for every authored declaration; `Check.lean` also prints definitions and theorem/instance types. The independent verifier inventories all compiled declarations and their transitive dependencies. Reports, logs, source identities and auxiliary code are included. Early failed author attempts retain their commands, exit codes and diagnostics, but their overwritten source versions were not preserved; contemporaneous source snapshots begin at final verification. Final-source replays are fully bound to their hashes. Local validation is distinct from maintainer acceptance.

References

- [1] The Last Math Competition, Conjecture 00000007744, [exact source at the submission base](#).
- [2] Mathlib v4.19.0, `SimpleGraph.AdjMatrix`, `Probability.Distributions.Uniform`, `Probability.Variance`, and `MeasureTheory.Measure.ProbabilityMeasure`. Exact pinned-source definition extracts are retained in the coordinator evidence.